

Hierarchical Fully Bayesian Approach

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Abstract—

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Each item corresponds to a driving segment or trajectory window τ_{ui} . We first map τ_{ui} to a feature vector through a deterministic feature extractor:

$$\mathbf{z}_{ui} = \phi(\tau_{ui}) \in \mathbb{R}^d.$$

For user u , we define the latent personalized reward function as

$$r_{ui} = r_u(\tau_{ui}) = \boldsymbol{\theta}_u^\top \mathbf{z}_{ui} = \boldsymbol{\theta}_u^\top \phi(\tau_{ui}).$$

The binary multi-user feedback $y_{ui} \in \{0, 1\}$ is labeled based on this personal reward. To express this observation process using an explicit equality form while maintaining the logistic choice structure, we introduce a latent variable formulation:

$$y_{ui} = \mathbb{I}(r_{ui} + \eta_{ui} > 0), \quad \eta_{ui} \sim \text{Logistic}(0, 1).$$

where $\mathbb{I}(\cdot)$ is the indicator function, and η_{ui} represents the random observation noise following a standard logistic distribution. By integrating out the latent logistic noise η_{ui} , this formulation can be equivalently expressed as an explicit choice probability form:

$$P(y_{ui} = 1 \mid \boldsymbol{\theta}_u, \mathbf{z}_{ui}) = \sigma(\boldsymbol{\theta}_u^\top \mathbf{z}_{ui}).$$

Consequently, the generative process for the binary feedback is modeled as a Bernoulli trial:

$$y_{ui} \mid \boldsymbol{\theta}_u, \mathbf{z}_{ui} \sim \text{Bernoulli}(\sigma(\boldsymbol{\theta}_u^\top \mathbf{z}_{ui})).$$

Users are heterogeneous but not unrelated. We model user-specific weights as draws from a shared population distribution,

$$\boldsymbol{\theta}_u \mid \boldsymbol{\mu}, \Sigma \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma), \quad u = 1, \dots, U.$$

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Here, $\boldsymbol{\mu}$ represents the population-average preference vector, while Σ represents variation and correlation in preference dimensions across users.

We place a Normal–Inverse–Wishart prior on the population parameters:

$$\Sigma \sim \text{IW}(\nu_0, \Lambda_0), \quad (1)$$

$$\boldsymbol{\mu} \mid \Sigma \sim \mathcal{N}\left(\mathbf{m}_0, \frac{1}{\kappa_0} \Sigma\right). \quad (2)$$

The target posterior is

$$p(\boldsymbol{\mu}, \Sigma, \boldsymbol{\theta}_{1:U} \mid \mathcal{D}_{1:U}),$$

where $\boldsymbol{\theta}_{1:U} = \{\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_U\}$.

A. User-Specific Reward Weight

For each user u , let

$$\mathbf{Z}_u = \begin{bmatrix} 1 \\ \mathbf{z}_{u1}^\top \\ \vdots \\ \mathbf{z}_{uN_u}^\top \end{bmatrix} \in \mathbb{R}^{N_u \times d}, \quad \mathbf{y}_u = (y_{u1}, \dots, y_{uN_u})^\top.$$

The Bernoulli-logistic likelihood is not conjugate to the Gaussian prior. To obtain conditionally Gaussian updates for $\boldsymbol{\theta}_u$, we use the Pólya–Gamma augmentation. Let $\psi_{ui} = \boldsymbol{\theta}_u^\top \mathbf{z}_{ui}$ and $c_{ui} = y_{ui} - 1/2$. Then the likelihood can be written as

$$p(y_{ui} \mid \psi_{ui}) \propto \exp(c_{ui} \psi_{ui}) \int_0^\infty \exp\left(-\frac{\omega_{ui} \psi_{ui}^2}{2}\right) p_{\text{PG}}(\omega_{ui} \mid 1, 0) d\omega_{ui}. \quad (3)$$

Conditioned on the current user weights,

$$\omega_{ui} \mid \boldsymbol{\theta}_u, \mathcal{D}_u \sim \text{PG}(1, \boldsymbol{\theta}_u^\top \mathbf{z}_{ui}).$$

For user u , define

$$\Omega_u = \text{diag}(\omega_{u1}, \dots, \omega_{uN_u}), \quad \mathbf{c}_u = \mathbf{y}_u - \frac{1}{2} \mathbf{1}_{N_u}.$$

Then

$$\boldsymbol{\theta}_u \mid \mathcal{D}_u, \boldsymbol{\omega}_u, \boldsymbol{\mu}, \Sigma \sim \mathcal{N}(\mathbf{m}_u, V_u),$$

with

$$V_u = (\mathbf{Z}_u^\top \Omega_u \mathbf{Z}_u + \Sigma^{-1})^{-1}, \quad (4)$$

$$\mathbf{m}_u = V_u (\mathbf{Z}_u^\top \mathbf{c}_u + \Sigma^{-1} \boldsymbol{\mu}). \quad (5)$$

B. Population Parameters

Conditioned on the current draws $\theta_1, \dots, \theta_U$, the Normal-Inverse-Wishart posterior is available in closed form [?]. Let

$$\bar{\theta} = \frac{1}{U} \sum_{u=1}^U \theta_u$$

and define

$$\begin{aligned} \kappa_U &= \kappa_0 + U, \\ \nu_U &= \nu_0 + U, \\ \mathbf{m}_U &= \frac{\kappa_0 \mathbf{m}_0 + U \bar{\theta}}{\kappa_0 + U}, \end{aligned} \quad (6)$$

and

$$\begin{aligned} \Lambda_U &= \Lambda_0 + \sum_{u=1}^U (\theta_u - \bar{\theta})(\theta_u - \bar{\theta})^\top \\ &\quad + \frac{\kappa_0 U}{\kappa_0 + U} (\bar{\theta} - \mathbf{m}_0)(\bar{\theta} - \mathbf{m}_0)^\top. \end{aligned} \quad (7)$$

The conditional distributions are

$$\Sigma \mid \theta_{1:U} \sim \text{IW}(\nu_U, \Lambda_U), \quad (8)$$

$$\mu \mid \Sigma, \theta_{1:U} \sim \mathcal{N}\left(\mathbf{m}_U, \frac{1}{\kappa_U} \Sigma\right). \quad (9)$$

C. New-User Personalization

For a new user $*$, the learned population posterior provides a personalized prior. For each posterior sample $(\mu^{(m)}, \Sigma^{(m)})$, we initialize

$$\theta_*^{(m)} \sim \mathcal{N}(\mu^{(m)}, \Sigma^{(m)}).$$

Given a small context set

$$\mathcal{D}_*^{\text{ctx}} = \{(\mathbf{z}_{*i}, y_{*i})\}_{i=1}^{N_*},$$

we update $\theta_*^{(m)}$ using the same Pólya–Gamma conditional Gaussian update:

$$V_*^{(m)} = \left(Z_*^\top \Omega_*^{(m)} Z_* + (\Sigma^{(m)})^{-1} \right)^{-1}, \quad (10)$$

$$\mathbf{m}_*^{(m)} = V_*^{(m)} \left(Z_*^\top \mathbf{c}_* + (\Sigma^{(m)})^{-1} \mu^{(m)} \right), \quad (11)$$

$$\theta_*^{(m)} \sim \mathcal{N}(\mathbf{m}_*^{(m)}, V_*^{(m)}).$$

For a query item with feature vector $\mathbf{z}$, the posterior predictive probability is approximated by Monte Carlo integration:

$$P(y = 1 \mid \mathbf{z}, \mathcal{D}_*^{\text{ctx}}, \mathcal{D}_{1:U}) \approx \frac{1}{M} \sum_{m=1}^M \sigma\left((\theta_*^{(m)})^\top \mathbf{z}\right). \quad (12)$$

The epistemic and aleatoric uncertainty of the prediction can be estimated by the posterior variance

$$\underbrace{\text{Var}(Y \mid \mathbf{z}, \mathcal{D})}_{\text{total predictive uncertainty}} = \underbrace{\mathbb{E}_m[p^{(m)}(1 - p^{(m)})]}_{\text{aleatoric}} + \underbrace{\text{Var}_m[p^{(m)}]}_{\text{epistemic}} \quad (13)$$

where $p^{(m)} = \sigma((\theta_*^{(m)})^\top \mathbf{z})$

V. NEXT PROGRESS

A. Feature Selection

The current framework assumes that the feature extractor $\phi(\tau_{ui})$ is predefined. However, including all available sensor-derived features may introduce redundant information, increase computational cost, and reduce interpretability. We therefore consider two complementary feature-selection criteria: (i) projection-predictive selection based on the Kullback–Leibler (KL) divergence from the full Bayesian model, and (ii) conditional predictive information gain based on the observed binary feedback.

The two criteria address different questions. The projection-predictive criterion identifies a compact subset that preserves the predictions of the full hierarchical model, whereas the conditional information-gain criterion identifies features that provide additional information about the observed feedback after conditioning on the features already selected. Neither criterion performs active query selection or attempts to reduce new-user parameter uncertainty.

1) *Selection Units*: Let $\mathcal{J} = \{1, \dots, d\}$ denote the set of individual feature indices and let $\mathcal{G} = \{1, \dots, G\}$ denote the set of sensors. For sensor g , define

$$\mathcal{J}_g = \{j \in \mathcal{J} : \text{feature } j \text{ is derived from sensor } g\}. \quad (14)$$

Thus, $\mathcal{J}_g$ may contain several trajectory-level statistics, such as the mean, variance, maximum, or temporal variation of the same sensor signal. The complete feature vector can be written as

$$\mathbf{z}_{ui} = [\mathbf{z}_{ui, \mathcal{J}_1}^\top, \dots, \mathbf{z}_{ui, \mathcal{J}_G}^\top]^\top. \quad (15)$$

We consider the following two sets of selectable units:

$$\mathcal{A}_{\text{sensor}} = \{\mathcal{J}_1, \dots, \mathcal{J}_G\}, \quad (16)$$

$$\mathcal{A}_{\text{feature}} = \{\{1\}, \dots, \{d\}\}. \quad (17)$$

Selecting $A = \mathcal{J}_g$ therefore selects all statistics extracted from sensor g , whereas selecting $A = \{j\}$ selects one individual sensor–statistic feature. The two selection procedures described below are applied separately at the sensor and individual-feature levels.

Let $S \subseteq \mathcal{J}$ denote the set of currently selected feature indices, and let $\mathbf{z}_{ui, S}$ denote the corresponding feature vector. The bias column, when included, is always retained and is not treated as a selectable unit. Its index set is denoted by S_0 .

2) *Projection-Predictive Selection*: The full hierarchical Bayesian model is first fitted using all available features. For posterior draw m , the reference predictive probability for observation (u, i) is

$$p_{ui}^{(m)} = \sigma\left((\theta_u^{(m)})^\top \mathbf{z}_{ui}\right). \quad (18)$$

These probabilities contain the predictive behavior learned by the full model, including both the population-level structure and user-specific preferences.

For a candidate subset S , we project each reference posterior draw onto a restricted logistic model using only $\mathbf{z}_{ui,S}$. The projected coefficient is defined as

$$\tilde{\boldsymbol{\theta}}_{u,S}^{(m)} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^{|S|}} \left\{ \frac{1}{N_u} \sum_{i=1}^{N_u} D_{\text{KL}} \left[\text{Bernoulli} (p_{ui}^{(m)}) \parallel \text{Bernoulli} (\sigma(\boldsymbol{\beta}^\top \mathbf{z}_{ui,S})) \right] + \frac{\lambda_{\text{proj}}}{2} \|\boldsymbol{\beta}_{\text{var}}\|_2^2 \right\}, \quad (19)$$

where $\boldsymbol{\beta}_{\text{var}}$ contains the coefficients associated with nonconstant feature columns. Constant columns, including the bias, are not penalized.

The projected predictive probability is then

$$q_{ui,S}^{(m)} = \sigma \left((\tilde{\boldsymbol{\theta}}_{u,S}^{(m)})^\top \mathbf{z}_{ui,S} \right). \quad (20)$$

For Bernoulli distributions, the pointwise KL divergence is

$$\begin{aligned} D_{\text{KL}} [\text{Bernoulli}(p) \parallel \text{Bernoulli}(q)] \\ = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}. \end{aligned} \quad (21)$$

Averaging over users, observations, and a set $\mathcal{M}_{\text{proj}}$ of retained posterior draws gives the projection loss

$$\mathcal{K}(S) = \frac{1}{|\mathcal{M}_{\text{proj}}| \sum_{u=1}^U N_u} \sum_{m \in \mathcal{M}_{\text{proj}}} \sum_{u=1}^U \sum_{i=1}^{N_u} D_{\text{KL}} [\text{Bernoulli} (p_{ui}^{(m)}) \parallel \text{Bernoulli} (q_{ui,S}^{(m)})]. \quad (22)$$

A smaller value of $\mathcal{K}(S)$ indicates that the restricted feature set more faithfully reproduces the predictions of the full Bayesian model.

Starting from the bias-only model, the next selectable unit is chosen by greedy forward selection:

$$A_t^* = \arg \min_{A \in \mathcal{A}, A \cap S_t = \emptyset} \mathcal{K}(S_t \cup A), \quad (23)$$

followed by

$$S_{t+1} = S_t \cup A_t^*, \quad S_0 = \{\text{bias}\}. \quad (24)$$

To define a stopping criterion, we measure the fraction of the bias-model projection loss removed by subset S :

$$R_{\text{KL}}(S) = 1 - \frac{\mathcal{K}(S)}{\mathcal{K}(S_0)}. \quad (25)$$

The smallest subset satisfying

$$R_{\text{KL}}(S) \geq \rho \quad (26)$$

is selected, where ρ denotes the desired predictive-information retention level. In the current implementation, $\rho = 0.95$ is used.

The projection in (??) is a deterministic soft-label logistic optimization. Consequently, Gibbs sampling is not repeated for every candidate subset. The observed feedback does not appear directly in the projection objective, but it affects the selection indirectly through the full-model posterior samples used in (??).

3) *Conditional Predictive Information Gain*: Projection-predictive selection measures agreement with the full model, but it does not directly quantify how much additional information a feature provides about the observed feedback. We therefore introduce a second criterion based on conditional mutual information.

Let U denote user identity. For a candidate unit A and a currently selected feature set S , we define

$$\begin{aligned} \mathcal{I}(A \mid S) &= I(Y; \mathbf{Z}_A \mid \mathbf{Z}_S, U) \\ &= H(Y \mid \mathbf{Z}_S, U) - H(Y \mid \mathbf{Z}_{S \cup A}, U). \end{aligned} \quad (27)$$

Equivalently,

$$\mathcal{I}(A \mid S) = \mathbb{E} \left[\log \frac{p(Y \mid \mathbf{Z}_{S \cup A}, U)}{p(Y \mid \mathbf{Z}_S, U)} \right]. \quad (28)$$

Thus, $\mathcal{I}(A \mid S)$ measures the additional predictive information provided by A after accounting for the selected features and user identity. Conditioning on U prevents between-user differences from being incorrectly attributed to sensor features.

The conditional probabilities in (??) are estimated using regularized user-conditioned logistic models. For subset S , the auxiliary conditional model has the form

$$\hat{p}_S(Y_{ui} = 1 \mid \mathbf{z}_{ui,S}, u) = \sigma(\alpha_u + \boldsymbol{\beta}_{u,S}^\top \mathbf{z}_{ui,S}), \quad (29)$$

where each user has a separate intercept and separate feature coefficients. The coefficients are estimated with an ℓ_2 regularization term.

To reduce in-sample optimism, the information gain is evaluated using within-user stratified cross-fitting. Let $k(ui)$ denote the validation fold containing observation (u, i) , and let $\hat{p}_S^{(-k(ui))}$ denote a conditional model fitted without that fold. Define the binary log score as

$$\ell(y, p) = y \log p + (1-y) \log(1-p). \quad (30)$$

The user-specific cross-fitted information gain is

$$\begin{aligned} \hat{\mathcal{I}}_u(A \mid S) &= \frac{1}{N_u} \sum_{i=1}^{N_u} \left[\ell(y_{ui}, \hat{p}_{S \cup A}^{(-k(ui))}) \right. \\ &\quad \left. - \ell(y_{ui}, \hat{p}_S^{(-k(ui))}) \right]. \end{aligned}$$

We assign equal weight to each user and define the overall estimate as

$$\hat{\mathcal{I}}(A \mid S) = \frac{1}{U} \sum_{u=1}^U \hat{\mathcal{I}}_u(A \mid S). \quad (31)$$

Its user-level standard error is estimated by

$$\widehat{\text{SE}}(A \mid S) = \left[\frac{1}{U(U-1)} \sum_{u=1}^U \left(\hat{\mathcal{I}}_u(A \mid S) - \hat{\mathcal{I}}(A \mid S) \right)^2 \right]^{1/2}. \quad (32)$$

Starting from the user-intercept-only model, the next selectable unit is chosen as

$$A_t^* = \arg \max_{A \in \mathcal{A}, A \cap S_t = \emptyset} \hat{\mathcal{I}}(A \mid S_t), \quad (33)$$

and the selected set is updated according to

$$S_{t+1} = S_t \cup A_t^*. \quad (34)$$

The forward search terminates when the best remaining candidate has nonpositive information gain,

$$\max_A \widehat{\mathcal{I}}(A | S_t) \leq \epsilon_{\text{IG}}, \quad (35)$$

where $\epsilon_{\text{IG}} = 0$ in the current implementation. A more conservative indicator of feature importance is

$$\widehat{\mathcal{I}}(A | S) - \widehat{\text{SE}}(A | S) > \epsilon_{\text{IG}}. \quad (36)$$

Unlike projection-predictive selection, the conditional information-gain criterion directly uses the observed feedback y_{ui} during both conditional-model fitting and held-out log-score evaluation. It should therefore be interpreted as a supervised, model-based estimate of conditional predictive information. Equality with the exact conditional mutual information in (??) requires correctly specified conditional probability models.

4) *Hierarchical Spike-and-Slab Selection*: The preceding selection criteria are applied after fitting a reference model or through an auxiliary predictive model. As an alternative, we incorporate feature selection directly into the hierarchical Bayesian model using binary inclusion variables.

Let $\mathcal{A} = \{A_1, \dots, A_K\}$ denote the selectable units defined previously. For individual-feature selection, A_g contains one feature index, whereas for sensor-level selection, A_g contains all statistics derived from sensor g . We associate each unit with a global binary inclusion indicator

$$\gamma_g \in \{0, 1\}, \quad g = 1, \dots, K. \quad (37)$$

The same γ_g is shared across users, so the resulting selection represents population-level relevance rather than user-specific selection.

Let $\tilde{\gamma} \in \{0, 1\}^d$ denote the feature-level mask obtained by expanding the unit-level indicators:

$$\tilde{\gamma}_j = \gamma_g, \quad j \in A_g. \quad (38)$$

When a bias term is included, its indicator is fixed as $\tilde{\gamma}_{\text{bias}} = 1$. Define

$$D_{\gamma} = \text{diag}(\tilde{\gamma}). \quad (39)$$

The effective user coefficient is then

$$\beta_u = D_{\gamma} \theta_u, \quad (40)$$

where θ_u denotes the latent slab coefficient satisfying the original hierarchical prior

$$\theta_u | \mu, \Sigma \sim \mathcal{N}(\mu, \Sigma). \quad (41)$$

Consequently, the binary-feedback model becomes

$$y_{ui} | \theta_u, \gamma, z_{ui} \sim \text{Bernoulli}(\sigma(z_{ui}^{\top} D_{\gamma} \theta_u)). \quad (42)$$

If $\gamma_g = 0$, every effective coefficient associated with unit A_g is exactly zero for all users. If $\gamma_g = 1$, the corresponding coefficients follow the hierarchical Gaussian slab distribution.

We place a Beta–Bernoulli prior on the inclusion indicators:

$$\pi \sim \text{Beta}(a_{\pi}, b_{\pi}), \quad (43)$$

$$\gamma_g | \pi \sim \text{Bernoulli}(\pi), \quad g = 1, \dots, K. \quad (44)$$

The parameter π represents the population-level proportion of included units. Its conditional posterior is

$$\pi | \gamma \sim \text{Beta}\left(a_{\pi} + \sum_{g=1}^K \gamma_g, b_{\pi} + K - \sum_{g=1}^K \gamma_g\right). \quad (45)$$

The Pólya–Gamma augmentation permits direct Gibbs updates of each inclusion indicator. Let

$$c_{ui,g} = z_{ui,A_g}^{\top} \theta_{u,A_g} \quad (46)$$

denote the contribution of candidate unit A_g , and define the linear predictor excluding this unit as

$$\eta_{ui,-g} = z_{ui}^{\top} D_{\gamma_{-g}} \theta_u + \epsilon_{ui}. \quad (47)$$

Here, $\epsilon_{ui} = 0$ when the additional observation-noise term is disabled. Since the augmented likelihood is proportional to

$$\exp\left(c_{ui} \psi_{ui} - \frac{\omega_{ui}}{2} \psi_{ui}^2\right), \quad c_{ui} = y_{ui} - \frac{1}{2}, \quad (48)$$

the conditional log-posterior odds of including unit A_g are

$$\Delta_g = \log \frac{\pi}{1 - \pi} + \sum_{u=1}^U \sum_{i=1}^{N_u} \left[c_{ui} c_{ui,g} - \frac{\omega_{ui}}{2} \{(\eta_{ui,-g} + c_{ui,g})^2 - \eta_{ui,-g}^2\} \right]. \quad (49)$$

Accordingly, the full conditional distribution is

$$\gamma_g | \text{rest} \sim \text{Bernoulli}(\sigma(\Delta_g)). \quad (50)$$

The indicators are updated sequentially within each Gibbs iteration, conditioning on the most recent values of the remaining indicators.

Given γ , define the masked design matrix

$$Z_{u,\gamma} = Z_u D_{\gamma}. \quad (51)$$

The conditional posterior of the slab coefficient remains Gaussian:

$$\theta_u | \omega_u, \gamma, \mu, \Sigma, \mathcal{D}_u \sim \mathcal{N}(\mathbf{m}_{u,\gamma}, V_{u,\gamma}), \quad (52)$$

where

$$V_{u,\gamma} = (Z_{u,\gamma}^{\top} \Omega_u Z_{u,\gamma} + \Sigma^{-1})^{-1}, \quad (53)$$

$$\mathbf{m}_{u,\gamma} = V_{u,\gamma} [Z_{u,\gamma}^{\top} (\mathbf{c}_u - \Omega_u \epsilon_u) + \Sigma^{-1} \mu]. \quad (54)$$

Thus, introducing γ preserves the conditionally conjugate Gaussian update for each user. The Normal–Inverse–Wishart updates of μ and Σ are unchanged and are applied to the latent slab coefficients $\theta_{1:U}$.

For retained posterior draw m , the effective coefficient used for prediction is

$$\beta_u^{(m)} = D_{\gamma^{(m)}} \theta_u^{(m)}. \quad (55)$$

For a new user, we first draw

$$\theta_*^{(m)} \sim \mathcal{N}(\mu^{(m)}, \Sigma^{(m)}) \quad (56)$$

and use

$$\beta_*^{(m)} = D_{\gamma^{(m)}} \theta_*^{(m)}. \quad (57)$$

During new-user personalization, the population-level indicator $\gamma^{(m)}$ remains fixed, while the corresponding slab coefficient $\theta_*^{(m)}$ is updated using the observed context.

The posterior inclusion probability of unit A_g is estimated as

$$\text{PIP}_g = P(\gamma_g = 1 \mid \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^M \gamma_g^{(m)}. \quad (58)$$

A unit with a high PIP_g is frequently included in posterior models. The subset

$$\hat{A}_{0.5} = \{A_g : \text{PIP}_g \geq 0.5\} \quad (59)$$

is reported as the median-probability model.

This method directly uses the observed feedback through the joint posterior and differs from the projection-predictive and conditional information-gain criteria. It modifies the hierarchical generative model rather than applying a separate post-hoc ranking criterion. However, correlated features may divide their posterior inclusion probabilities, and the binary indicators can exhibit slow switching. Therefore, posterior inclusion probabilities should be interpreted together with the indicator traces and the number of posterior transitions. Moreover, because γ is shared across users, the current formulation may suppress a feature that is important only for a small subset of users.

5) *Interpretation and Limitations:* The two criteria provide complementary interpretations. A low $\mathcal{K}(S)$ indicates that subset S preserves the predictive behavior of the full hierarchical Bayesian model, while a large $\hat{\mathcal{I}}(A \mid S)$ indicates that candidate A provides additional information about the observed feedback beyond the features already selected.

Both methods use greedy forward selection and therefore do not guarantee a globally optimal subset. In addition, highly correlated features may substitute for one another, so a feature assigned low incremental importance is not necessarily uninformative in isolation. At the sensor level, groups containing more derived statistics also have greater representational capacity than smaller groups. Finally, the conditional information-gain estimator evaluates within-user predictive relevance; it does not evaluate generalization to an entirely unseen user.

Accordingly, the projection-predictive and conditional-information procedures are reported separately as two feature-selection analyses. The resulting subsets should be treated as candidate reduced representations and subsequently verified by refitting and evaluating the hierarchical model on held-out data.

B. Add Bias Term

C. Scalable Bayesian Reward Learning

VI. EXPERIMENTS

VII. CONCLUSION AND FUTURE WORK

REFERENCES

Algorithm 1 Population-Level Bayesian Reward Learning

Require: Multi-user data $\mathcal{D}_{1:U}$, NIW hyperparameters, burn-in B , retained samples M

Ensure: Posterior samples $\{\mu^{(m)}, \Sigma^{(m)}, \theta_{1:U}^{(m)}\}_{m=1}^M$

- 1: Extract and standardize features $\mathbf{z}_{ui} = \phi(\tau_{ui})$ for all users.
 - 2: Initialize μ , Σ , and $\{\theta_u\}_{u=1}^U$.
 - 3: Set $s \leftarrow 0$.
 - 4: **for** $t = 1$ to $B + M$ **do**
 - 5: **for** $u = 1$ to U **do**
 - 6: Sample $\omega_{ui} \sim \text{PG}(1, \theta_u^\top \mathbf{z}_{ui})$ for $i = 1, \dots, N_u$.
 - 7: Sample $\theta_u \mid \mathcal{D}_u, \omega_u, \mu, \Sigma$ using (??)–(??).
 - 8: **end for**
 - 9: Compute NIW posterior parameters using (??)–(??).
 - 10: Sample $\Sigma \mid \theta_{1:U}$ using (??).
 - 11: Sample $\mu \mid \Sigma, \theta_{1:U}$ using (??).
 - 12: **end for**
 - 13: **return** $\{\mu^{(m)}, \Sigma^{(m)}, \theta_{1:U}^{(m)}\}_{m=1}^M$
-

Algorithm 2 New-User Personalization and Prediction

Require: Population posterior samples $\{\mu^{(m)}, \Sigma^{(m)}\}_{m=1}^M$, context data $\mathcal{D}_*^{\text{ctx}}$, query features $\{\mathbf{z}_j\}_{j=1}^{N_q}$, adaptation sweeps L

Ensure: Predictive probabilities $\{\hat{p}_j\}_{j=1}^{N_q}$

- 1: **for** $m = 1$ to M **do**
 - 2: Initialize $\theta_*^{(m)} \sim \mathcal{N}(\mu^{(m)}, \Sigma^{(m)})$.
 - 3: **end for**
 - 4: **for** $\ell = 1$ to L **do**
 - 5: **for** $m = 1$ to M **do**
 - 6: Sample PG variables $\omega_{*i}^{(m)} \sim \text{PG}(1, (\theta_*^{(m)})^\top \mathbf{z}_{*i})$.
 - 7: Update $\theta_*^{(m)}$ using the conditional Gaussian update in the new-user equations.
 - 8: **end for**
 - 9: **end for**
 - 10: **for** $j = 1$ to N_q **do**
 - 11: Compute $\hat{p}_j$ using (??).
 - 12: Estimate epistemic uncertainty by the posterior variance of $\{\sigma((\theta_*^{(m)})^\top \mathbf{z}_j)\}_{m=1}^M$.
 - 13: **end for**
 - 14: **return** $\{\hat{p}_j\}_{j=1}^{N_q}$
-